

## Stern-Gerlach 3

From the previous sections we have

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

where

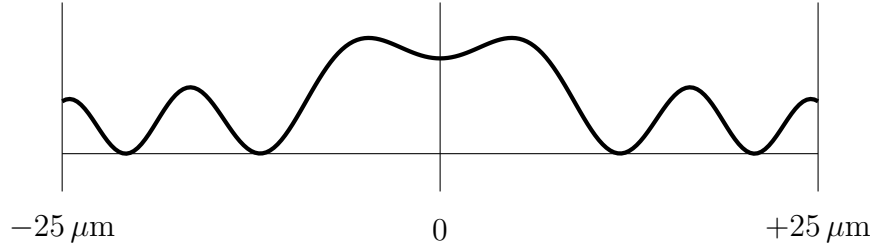
$$\begin{aligned} \psi_1 &= \text{Ai} \left[ \left( \frac{\alpha e}{2\hbar} \right)^{1/3} \frac{\alpha e \hbar}{8m^2} t^2 + \left( \frac{\alpha e}{2\hbar} \right)^{1/3} z \right] \exp \left( -\frac{ie\alpha z t}{4m} \right) \exp \left( -\frac{ieB_0 t}{2m} \right) \\ \psi_2 &= \text{Ai} \left[ \left( \frac{\alpha e}{2\hbar} \right)^{1/3} \frac{\alpha e \hbar}{8m^2} t^2 - \left( \frac{\alpha e}{2\hbar} \right)^{1/3} z \right] \exp \left( +\frac{ie\alpha z t}{4m} \right) \exp \left( +\frac{ieB_0 t}{2m} \right) \end{aligned}$$

The probability density for silver atoms is

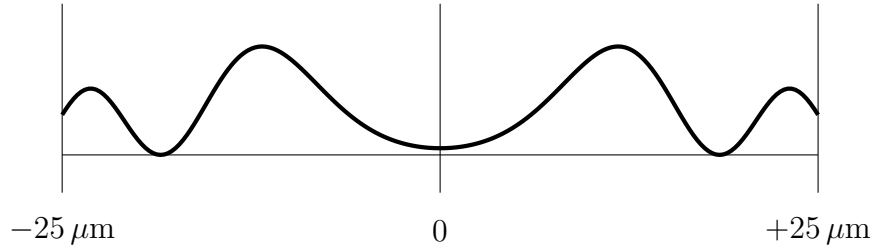
$$f(t, z) = \Psi^\dagger \Psi = |\psi_1|^2 + |\psi_2|^2$$

The following plots show the time evolution of silver atom density. Because we are using a polynomial approximation for the Airy function, it is necessary to use small values of deflection, i.e.,  $25 \mu\text{m}$ .

$t = 0$



$t = 0.1$



$t = 0.15$

